GRAPHS OF CONTINUOUS BUT NON-AFFINE FUNCTIONS ARE NEVER SELF-SIMILAR

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Abstract. Bandt and Kravchenko [2] proved if a self-similar set spans $\mathbb{R}^n$, then there is no tangent hyperplane at any point of the set. In particular, this indicates that a smooth planar curve is self-similar if and only if it is a straight line. When restricting curves to graphs of continuous functions, we can show that the graph of a continuous function is self-similar if and only if the graph is a straight line i.e., the underlying function is affine.

1. Introduction

Recall that a map $S : \mathbb{R}^n \to \mathbb{R}^n$ is said to be a *similitude* (e.g., [4, 5]) if $S(x) = rOx + b$, where $r \in (0, 1)$, $b \in \mathbb{R}^n$, and O is an orthogonal matrix. A compact set $K \subset \mathbb{R}^n$ is *self-similar* if there are similitudes $\{S_i\}_{i=1}^m$, such that

$$K = \bigcup_{i=1}^{m} S_i(K). \tag{1.1}$$

The structure of self-similar sets become relatively well-understood when it satisfies the open set condition (**OSC**) introduced by Hutchinson in [7]. Here, the OSC is satisfied if there exists a nonempty open set $V \subset \mathbb{R}^n$ such that $\bigcup_{i=1}^m S_i(V) \subset V$ and $S_i(V) \cap S_j(V) = \emptyset$ for $i \neq j$. In this case, $\dim_{\mathrm{H}} K = s$, where s is the unique solution of $\sum_{j=1}^k r_i^s = 1$. Another characteristic of the OSC can be found in [1] by Bandt and Graf.

It is widely acknowledged that fractals are inherently non-smooth. Yet, there has been limited exploration into geometric objects that exhibit both self-similarity and smoothness. However, Bandt and colleagues have been pioneers in this field, achieving various significant results. For instance, Bandt and Mubarak [3] established that any differentiable subcurve of the classical Sierpinski carpet must be a line segment. It is worth pointing out that Bandt and Kravchenko [2] demonstrated that a self-similar set spanning $\mathbb{R}^n$ cannot possess a tangent hyperplane at any point within the set, a finding with broad applications. For example, it suggests that a self-similar planar curve, if differentiable at some point, can only be a straight line.

In the present study, we concern with a special class of curves: the graphs of continuous functions, i.e.,

$$G := \{(x, y) \in \mathbb{R}^2 : y = f(x), x \in I\},$$

where f is continuous on I and I is a closed and bounded interval. We notice that many continuous but nowhere differentiable functions exhibit high "self-similarity" in their graphs. Notable examples include Weierstrass's and Takagi's function function (e.g., see [6, 8]), as illustrated in Figure 1.1 and Figure 1.2 respectively. However, because these functions are

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not smooth, it's hard to tell if they are self-similar by using existing results on smooth self-similar sets given by Bandt and Kravchenko [2]. This complexity has spurred our interest in developing new methodologies to address such questions.

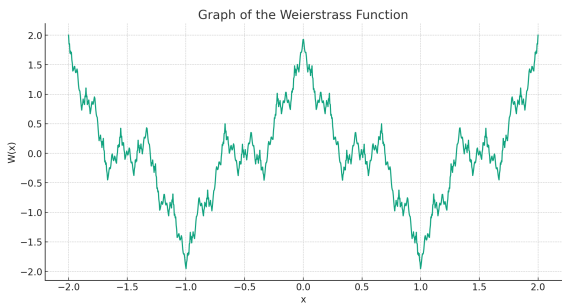

FIGURE 1.1. The graph of Weierstrass's function

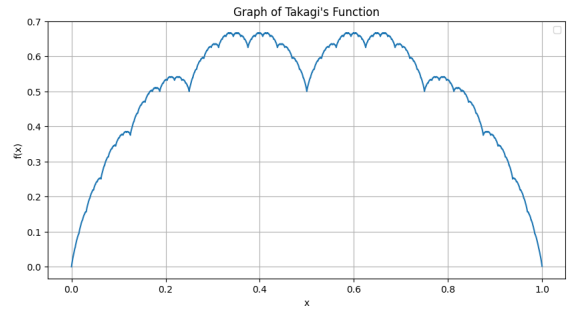

FIGURE 1.2. The graph of Takagi's function

In this work, we concern about which geometric shapes can be realized as a self-similar set. In particular, we propose the following problem for planar graphs associated with one real variable continuous functions:

When is the planar graph of a continuous function self-similar?

This originates from the question proposed by the second author and Jiang: Can the graph of a C^1 and non-affine function f on a closed and bounded interval be self-similar? This is discussed in [9, Question 3.1]. In addition, Bandt and Kravchenko's research reveals the absence of tangent spaces in self-similar curves (see [2, Theorem 1]), suggesting the C^1 regularity seems to be overly stringent for the graphs of such functions.

In this article, we are aiming to give an affirmative and comprehensive answer to the above question. To be more precise, our main result is stated as follows.

Theorem 1.1. *Let I be a closed and bounded interval, $f : I \to \mathbb{R}$ be a continuous function and $G = \{(x, f(x)) : x \in I\}$. Then the following two statements are equivalent:*

- *G is a self-similar set;*
- *The underlying function $f(x) = ax + b$. In other words, f is an affine function on I.*

It is worth remarking that due to the existence of continuous self-similar planar curves, none of hypotheses in Theorem 1.1 can be weaken. Moreover, Theorem 1.1 strengthens [9, Corollary 1.6] by the second author and Jiang. According to Theorem 1.1, graphs of continuous functions that are non-affine, are not self-similar. In particular, the graphs of Weierstrass's function, Takagi's function and Cantor-Lebesgue's function etc are not self-similar, see Corollary 3.9.

The paper is organized as follows. In Section 2, we recall some notations and review some preliminaries. In Section 3, we prove our main Theorem 1.1. In subsection 3.1, we explain the strategy of the proof. The proof consists of three steps, occupy subsections 3.2, 3.3 and 3.4 respectively.

2. PREPARATION

2.1. **Notation and Preliminaries.** We use the convention that $\mathbb{N} := \{1, 2, 3, \dots\}$, $\mathbb{N}_0 := \{0\} \cup \mathbb{N}$, $\mathbb{Z}$ the set of integers, and $\mathbb{Q}$ the set of all rational numbers. For a set A, the

cardinality of A is denoted by $\#A$. For a $k \in \mathbb{N}$, define $[k] := \{1, \dots, k\}$. For a $l \in \mathbb{N}$, define

$$[k]^l := \{(x_1, \dots, x_l) : x_i \in [k],\ i = 1, \dots, l\}.$$

For a continuous function f on a closed and bounded interval I, the *graph* of f, denoted as G, is defined by $G := \{(x, f(x)) \in \mathbb{R}^2 : x \in I\}$.

Next, let us discuss more notations for self-similar set on $\mathbb{R}^2$. We say a map $S : \mathbb{R}^2 \to \mathbb{R}^2$ is a *similitude* if $S(x) = r\rho_\theta x + b$ with $r \in (0,1), b \in \mathbb{R}^2$ and $\rho_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$. For our purpose, (1.1) reduces to

$$G = \bigcup_{i=1}^{k} S_i(G).$$

where $S_i(x) = r_i\rho_{\theta_i}(x) + b_i$. Moreover, for a non-negative integer l and $\alpha = (i_1, \dots, i_l) \in [k]^l$, define $S_\alpha := S_{i_l} \circ \cdots \circ S_{i_1}$ and $r_\alpha := r_{i_l} \cdots r_{i_1}$.

In addition, denote by the unit circle centred at the origin $S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, endowed with the circle metric. Call $(0, 1)$ the North Pole $\mathcal{N}$ of S^1, and $(0, -1)$ the South Pole $\mathcal{S}$ of S^1.

Finally, let's review some results for a dynamical system on S^1. Let us regard $\rho_\theta : S^1 \to S^1$ as $e^{2\pi i x} \mapsto e^{2\pi i(x+\alpha)}$, with $\alpha = \frac{\theta}{2\pi}$. For the sake of completeness, we provies the details of the proof.

Proposition 2.1. *If $\alpha \in \mathbb{Q}^c$, then the set $\{\rho_\theta^k(\xi)\}_{k=1}^\infty = \{e^{2\pi i(x+k\alpha)}\}_{k=1}^\infty$ is dense in S^1 for any $\xi = e^{2\pi i x} \in S^1$.*

Proof of Proposition 2.1. By symmetry, without loss of generality, we assume $\xi = e^{2\pi i 0} = 1$ throughout the proof. It suffices to show that for any $\zeta \in S^1$ and any $\epsilon > 0$, there exists $s \in \mathbb{N}$, such that

$$|\rho_\theta^s(1) - \zeta| = |e^{2\pi i s\alpha} - \zeta| < \epsilon. \tag{2.1}$$

By the hypothesis $\alpha \in \mathbb{Q}^c$, it directly yields that $e^{2\pi i m\alpha} \neq e^{2\pi i n\alpha}$, whenever integers $m \neq n$. Next, note that S^1 is of finite length. We claim that there exists integers $m < n$, such that

$$|e^{2\pi i m\alpha} - e^{2\pi i n\alpha}| < \epsilon.$$

To see this, suppose on contrary that any two points in $\{e^{2\pi i k\alpha}\}_{k=1}^\infty$ are at least ϵ (in the circle distance) away from each other. As each point in $\{e^{2\pi i k\alpha}\}_{k=1}^\infty$ are mutually different, there will be N points in $\{e^{2\pi i k\alpha}\}_{k=1}^\infty$ on S^1 are mutually at least distance least ϵ away from each other. This implies the length of S^1 would be at least of $N\epsilon$. Since N is arbitrary, this is obviously impossible. Thus we complete the proof of the claim.

Then $|\rho_\theta^{(i+1)(n-m)}(1) - \rho_\theta^{i(n-m)}(1)| = |e^{2\pi i(n-m)\alpha} - 1| = |e^{2\pi i m\alpha} - e^{2\pi i n\alpha}| < \epsilon$. Hence, there exists $s_0 \in \mathbb{Z}$ such that $|\rho_\theta^{s_0(n-m)}(1) - \zeta| < \epsilon$. Here $s_0(n-m)$ is the desired s in (2.1). Thus, we complete the proof of Proposition 2.1.

□

Deducing from Proposition 2.1, we have

Corollary 2.2. *If I is an interval with nonempty interior in S^1 and $\alpha \in \mathbb{Q}^c$, then*

$$\bigcup_{i=1}^{\infty} \rho_\theta^i(I) = S^1.$$

Proof of Corollary 2.2. By the hypothesis that the interval I has nonempty interior, the length of I equals to $l > 0$. Set $\tau = e^{2\pi i x}$ be the center of I. By Proposition 2.1, the set $\{\rho_\theta^k(\xi)\}_{k=1}^\infty$ is dense in S^1. Therefore, for any point $\zeta \in S^1$, there is a positive integer k_0, such that

$$|\rho_\theta^{k_0}(\tau) - \zeta| = |e^{2\pi i(x+k_0\alpha)} - \zeta| < \frac{l}{2}.$$

In other word, $\rho_\theta^{k_0}(I)$ contains ζ. By the arbitrariness of ζ, this directly yields that $\bigcup_{i=1}^{\infty} \rho_\theta^i(I) = S^1$, which completes the proof of Corollary 2.2. □

3. Proof of Theorem 1.1

3.1. **Strategy of the proof.** Let's discuss the strategy of the proof of Theorem 1.1 as follows. The main technical challenge to overcome in the proof of Theorem 1.1 lies in dealing with the non-differentiability (only continuity) of the function. Our proof is independent of any techniques based on curvature or other differentiable tools, and is quite different from [9, Theorem 1.5].

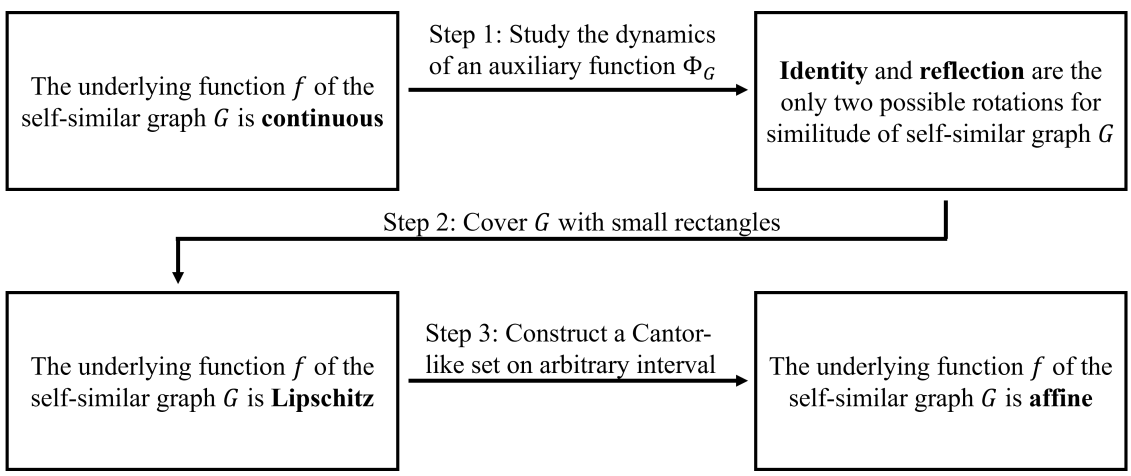

Figure 3.1. The strategy to show the progressive properties of the underlying function of a self-similar graph G.

In fact, as illustrated in Figure 3.1, our proof is decomposed into three steps, each elucidating further properties of the underlying function f of a self-similar graph G. The progression from continuity, through Lipschitz continuity, to affine nature, is visually summarized and demonstrates the logical flow of the argument.

Step 1: Firstly, we construct an auxiliary map Φ_G (see (3.1)). By studying the dynamic of $\operatorname{Im}\Phi_G$ under rotation that is associated with a similitude on G (see Lemma 3.3, Lemma 3.4, and Corollary 3.6), we conclude that the only possible rotations for similitutes G are identity and reflection (see Proposition 3.1).

Step 2: Next, by Step 1 we can frame G by a minimal rectangle R and focus on the iteration of R under similitudes. This reduces the small pieces of G to their projection on x-axis. Then by a covering argument (see Proposition 3.7), it enables us to show that the underlying function f is Lipschitz continuous.

Step 3: Finally, given any subinterval $[a, b]$ in $[0, 1]$, we can find a subinterval $[s, t]$ where $\frac{f(t)-f(s)}{t-s} = f(1)-f(0)$ (see Lemma 3.8). Applying Lemma 3.8 repeatedly, we construct a Cantor-like subset in $[a, b]$. Combined with the Lipschitz continuity of f obtained at Step 2, we then take a distortion estimate for f on the Cantor-like set, and conclude that f must be affine on $[0, 1]$, as we want.

In this section, we aim to prove Theorem 1.1. Due to the observation that a self-similar set remains self-similar after rescaling and translation, we can reduce the hypothesis of a closed and bounded interval to $[0, 1]$ in the proof of Theorem 1.1. Therefore, without loss of generality, we always assume $I = [0, 1]$ throughout the rest part of this section.

The proof of Theorem 1.1 is divided into three subsections, namely Subsection 3.2, 3.3 and 3.4. Each subsection corresponds to a distinct step illustrated in Figure 3.2. Proof of Theorem 1.1 will be given at the end of Subsection 3.4.

3.2. **Restriction on Possible Rotations for Similitudes of the Graph.** In this subsection, we will realize Step 1 in Figure 3.1, specifically, to demonstrate that there are only two possible rotations for the similitudes of the self-similar graph (see Proposition 3.1). To achieve this, we construct an auxiliary map Φ_G for the self-similar graph G (see (3.1)). There are several useful properties of Φ_G (see Lemma 3.3, Lemma 3.4, and Corollary 3.6). By combining all these properties and considering the dynamics of $\operatorname{Im}\Phi_G$ under rotation, we deduce our main result for this subsection as follows.

Proposition 3.1. *Let G be the graph of a continuous function f on $[0, 1]$, and similitude $S(x) = r\rho_\theta(x) + b$. If $S(G) \subset G$, then $\rho_\theta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.*

To prove Proposition 3.1, we construct an auxiliary map Φ as follows. Let G be the graph of continuous function f on $[0, 1]$. Define

$$\Delta_G := \{(\overrightarrow{a_1}, \overrightarrow{a_2}) \in G \times G : \overrightarrow{a_1} \neq \overrightarrow{a_2}\}.$$

Define

$$\Phi_G : \Delta_G \to S^1,\ (\overrightarrow{a_1}, \overrightarrow{a_2}) \mapsto \frac{\overrightarrow{a_2} - \overrightarrow{a_1}}{\|\overrightarrow{a_2} - \overrightarrow{a_1}\|}. \tag{3.1}$$

Lemma 3.2. *The map Φ_G has the following properties:*

(1) $\Phi_G(\overrightarrow{a_2}, \overrightarrow{a_1}) = -\Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2})$;
(2) $\mathcal{N}, \mathcal{S} \notin \operatorname{Im}\Phi_G$;
(3) $\operatorname{Im}\Phi_G$ *consists of two connected components.*

Proof of Lemma 3.2.

(1) Property (1) is obvious from the definition of Φ_G in (3.1).
(2) We prove Property (2) by contradiction. Suppose that $\mathcal{N} \in \operatorname{Im}\Phi_G$. Then there exists $\overrightarrow{a_1} \neq \overrightarrow{a_2}$ in G such that $\Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}) = \mathcal{N}$. Write $\overrightarrow{a_1} = (t_1, f(t_1))$ and $\overrightarrow{a_2} = (t_2, f(t_2))$ for some $t_1, t_2 \in [0, 1]$. Since $\overrightarrow{a_1} \neq \overrightarrow{a_2}$, we have $t_1 \neq t_2$. However, $\Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}) = \mathcal{N}$ implies $t_1 = t_2$, which is a contradiction. A similar argument applies to $\mathcal{S}$. Thus we conclude the proof of the second property.
(3) For Property (3), we define a homeomorphism $F : t \mapsto (t, f(t))$. Then $(F, F)|_{\Delta_I}$ is a homeomorphism between $\Delta_I := \{(t_1, t_2) \in [0, 1] \times [0, 1] : t_1 \neq t_2\}$ and Δ_G. Notice that Δ_I consists of exactly two connected regions, say Δ_I^1 and Δ_I^2. Thus, Δ_G also consists of two connected regions $\Delta_G^1 = (F, F)(\Delta_I^1)$ and $\Delta_G^2 = (F, F)(\Delta_I^2)$. As Φ_G is

continuous, $\Phi_G(\Delta_G^1)$ and $\Phi_G(\Delta_G^2)$ are both connected. Notice they consist of $\operatorname{Im}\Phi_G$. This completes the proof of the third property.

□

The following lemma will be useful later in the proof of Proposition 3.1.

Lemma 3.3. *Let G be the graph of continuous function f on $[0,1]$. Then f is affine if and only if $\operatorname{Im}\Phi_G$ consists of exactly two antipodal points. Here, two points $x, y \in S^1$ are antipodal if $x = -y$.*

Proof of Lemma 3.3. " $\Rightarrow$: " Suppose f is affine, say $f(x) = kx + b$ for some $k, b \in \mathbb{R}$ and $k \neq 0$. Then for any $(\overrightarrow{a_1}, \overrightarrow{a_2}) \in \Delta_G$, writing $\overrightarrow{a_1} = (a_1, kt_1 + b)$ and $\overrightarrow{a_2} = (t_2, kt_2 + b)$, we have:

$$\Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}) = \frac{(t_2 - t_1, k(t_2 - t_1))}{\|(t_2 - t_1, k(t_2 - t_1))\|} = \begin{cases} (\frac{1}{\sqrt{1+k^2}}, \frac{k}{\sqrt{1+k^2}}), & \text{if } t_1 < t_2; \\ -(\frac{1}{\sqrt{1+k^2}}, \frac{k}{\sqrt{1+k^2}}), & \text{otherwise.} \end{cases} \tag{3.2}$$

Therefore, $\operatorname{Im}\Phi_G = \left\{(\frac{1}{\sqrt{1+k^2}}, \frac{k}{\sqrt{1+k^2}}), -(\frac{1}{\sqrt{1+k^2}}, \frac{k}{\sqrt{1+k^2}})\right\}$.

" $\Leftarrow$: " Conversely, assume $\operatorname{Im}\Phi_G$ consists of two antipodal points. Write

$$\operatorname{Im}\Phi_G = \{(\cos\phi, \sin\phi), -(\cos\phi, \sin\phi)\}$$

for some $\phi \in (-\pi, \pi)$. Consequently, for all $x \in (0, 1]$, we have

$$\Phi_G((0, f(0)), (x, f(x))) = (\cos\phi, \sin\phi).$$

This implies $f(x) - f(0) = \tan\phi \cdot x$ for all $x \in [0, 1]$. In other word, f is affine. □

Next, we have

Lemma 3.4. *Let G be the graph of continuous function f on $[0,1]$. If a similitude $S(x) = r \cdot \rho_\theta(x) + b$ satisfies $S(G) \subset G$, then $(S(\overrightarrow{a_1}), S(\overrightarrow{a_2})) \in \Delta_G$ for all $(\overrightarrow{a_1}, \overrightarrow{a_2}) \in \Delta_G$ and*

$$\Phi_G(S(\overrightarrow{a_1}), S(\overrightarrow{a_2})) = \rho_\theta \Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}). \tag{3.3}$$

This implies that the diagram in Figure 3.2 commutes.

$$\begin{tikzcd} \Delta_G \arrow[r, "\Phi_G"] \arrow[d, "S\times S"'] & S^1 \arrow[d, "\rho_\theta"] \\ \Delta_G \arrow[r, "\Phi_G"] & S^1 \end{tikzcd}$$

FIGURE 3.2. The commutative diagram

Proof of Lemma 3.4. For the first assertion, we proceed the proof by contradiction. Suppose that $(S(\overrightarrow{a_1}), S(\overrightarrow{a_2})) \notin \Delta_G$. Since $S(\overrightarrow{a_1}), S(\overrightarrow{a_2}) \in S(G) \subset G$, we must have $S(\overrightarrow{a_1}) = S(\overrightarrow{a_2})$. However, invertibility of S implies $\overrightarrow{a_1} = \overrightarrow{a_2}$. This contradicts to the assumption that $(\overrightarrow{a_1}, \overrightarrow{a_2}) \in \Delta_G$. This completes the proof of the first assertion.

For the second assertion, we observe that

$$\begin{aligned}
\Phi_G(S(\overrightarrow{a_1}), S(\overrightarrow{a_2})) &= \frac{r \cdot \rho_\theta(\overrightarrow{a_2}) + b - (r \cdot \rho_\theta(\overrightarrow{a_1}) + b)}{\|r \cdot \rho_\theta(\overrightarrow{a_2}) + b - (r \cdot \rho_\theta(\overrightarrow{a_1}) + b)\|} && \text{(by (3.1))} \\
&= \frac{r}{|r|} \cdot \frac{\rho_\theta(\overrightarrow{a_2}) - \rho_\theta(\overrightarrow{a_1})}{\|\rho_\theta(\overrightarrow{a_2}) - \rho_\theta(\overrightarrow{a_1})\|} \\
&= \frac{\rho_\theta(\overrightarrow{a_2} - \overrightarrow{a_1})}{\|\overrightarrow{a_2} - \overrightarrow{a_1}\|} && \text{(as } \rho_\theta \text{ is orthogonal and } r \in (0,1)) \\
&= \rho_\theta \Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}), && \text{(by (3.1))}
\end{aligned}$$

This completes the proof of the second assertion. □

Remark 3.5. It is noteworthy that the commutativity of the diagram is independent of the values of r and b in the similitude S. Meanwhile, due to (3.3), it follows that $\operatorname{Im}\Phi_G$ is invariant under ρ_θ.

Corollary 3.6. *Let $S(G) \subset G$ for a similitude S. Then, the following two statements hold:*

(1) *For any $m \in \mathbb{N}$ and $(\overrightarrow{a_1}, \overrightarrow{a_2}) \in \Delta_G$, we have $(S^m(\overrightarrow{a_1}), S^m(\overrightarrow{a_2})) \in \Delta_G$ and*

$$\Phi_G(S^m(\overrightarrow{a_1}), S^m(\overrightarrow{a_2})) = \rho_\theta^m \Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}). \tag{3.4}$$

(2) *For any subset $J \subset \operatorname{Im}(\Phi_G)$, we have*

$$\Phi_G(S^m(\Phi_G^{-1}(J))) = \rho_\theta^m(J), \tag{3.5}$$

where S acts on Δ_G by assigning $(\overrightarrow{a_1}, \overrightarrow{a_2})$ to $(S(\overrightarrow{a_1}), S(\overrightarrow{a_2}))$.

Proof of Corollary 3.6.

(1) Applying Lemma 3.4 m times yields the first assertion.

(2) For the second assertion, we show that sets on both sides of (3.5) are mutually inclusive.

For a point $p \in \Phi_G(S^m(\Phi_G^{-1}(J)))$, there exists a pair $(\overrightarrow{a_1}, \overrightarrow{a_2}) \in \Delta_G$ such that $\Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}) \in J$ and $p = \Phi_G(S^m(\overrightarrow{a_1}), S^m(\overrightarrow{a_2}))$. Due to (3.4), we have

$$\Phi_G(S^m(\overrightarrow{a_1}), S^m(\overrightarrow{a_2})) = \rho_\theta^m \Phi_G(\overrightarrow{a_1}, \overrightarrow{a_2}) \in \rho_\theta^m(J).$$

This indicates that $\Phi_G(S^m(\Phi_G^{-1}(J))) \subset \rho_\theta^m(J)$.

Conversely, for a point $p \in \rho_\theta^m(J)$, we have $\rho_\theta^{-m}(p) \in J \subset \operatorname{Im}\Phi_G$. Then there exists $(a_1, a_2) \in \Delta_G$ such that $\rho_\theta^{-m}(p) = \Phi_G(a_1, a_2)$. In other word, $p = \rho_\theta^m \circ \Phi_G(a_1, a_2)$. On the other hand, $\rho_\theta^m \circ \Phi_G(a_1, a_2) = \Phi_G(S^m(a_1), S^m(a_2))$ by (3.4). This implies $\rho_\theta^m(J) \subset \Phi_G(S^m(\Phi_G^{-1}(J)))$

This finishes the proof of the second assertion of Corollary 3.6.

□

We are now turning to the proof of Proposition 3.1.

Proof of Proposition 3.1. We prove the Proposition 3.1 in two cases: when f is affine and when f is non-affine.

Firstly, suppose f is affine. According to Lemma 3.3, it follows that $\operatorname{Im}\Phi_G$ consists of exactly two antipodal points. By Remark 3.5, $\operatorname{Im}\Phi_G$ is invariant under ρ_θ. On the other hand, the rotations that preserve two antipodal points are exactly $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. This completes the proof of the affine case.

Secondly, suppose f is non-affine, then we can choose an interval $J \subset \operatorname{Im}\Phi_G$ on S^1 with non-empty interior by the third assertion in Lemma 3.2 and Lemma 3.3. According to (3.5), we have $\rho_\theta^i(J) \subset \operatorname{Im}\Phi_G$ for any $i \in \mathbb{N}_0$. Consequently, it follows that

$$\bigcup_{i=0}^{n} \rho_\theta^i(J) \subset \operatorname{Im}\Phi_G \text{ for any } n \in \mathbb{N}_0 \tag{3.6}$$

and

$$\bigcup_{i=0}^{\infty} \rho_\theta^i(J) \subset \operatorname{Im}\Phi_G. \tag{3.7}$$

We divide the remaining part of the proof into two cases: when $\theta/2\pi$ is irrational and $\theta/2\pi$ is rational.

Suppose $\theta/2\pi \in \mathbb{Q}^c$. Then $\bigcup_{m=1}^{\infty} \rho_\theta^m(J)$ covers the entire circle S^1 by Corollary 2.2. Together with (3.7), this implies that $\operatorname{Im}\Phi_G = S^1$. This contradicts to the fact that $\mathcal{N}$ and $\mathcal{S}$ are not in $\operatorname{Im}\Phi_G$, as stated in the second assertion of Lemma 3.2. In other word, $\theta/2\pi$ must be rational.

Suppose $\theta/2\pi \in \mathbb{Q}$. We can write $\theta/2\pi = m/n$ with $m \in \mathbb{Z}, n \in \mathbb{N}$ and $\gcd(m,n) = 1$.

For $n \geq 3$, denote by $\mathcal{N}_i = \rho_\theta^i(\mathcal{N})$, $\mathcal{S}_i = \rho_\theta^i(\mathcal{S})$ and $J_i = \rho_\theta^i(J)$ for $i = 0, \ldots, n-1$. Thus, S^1 is partitioned into $2n$ segments by $2n$ points $\mathcal{N}_0, \ldots, \mathcal{N}_{n-1}, \mathcal{S}_0, \ldots, \mathcal{S}_{n-1}$. The intervals $J_0, \ldots, J_{n-1}$ fall into n different segments. Furthermore, any two adjacent segments contain exactly one J_l for some l, as pictured in Figure 3.3.

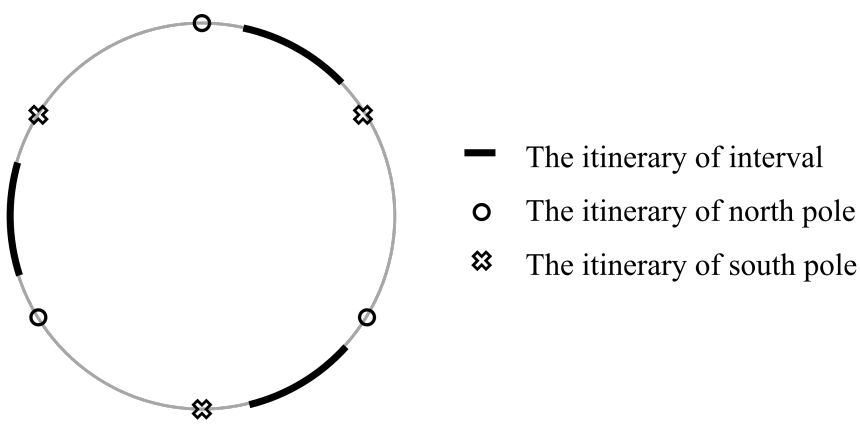

FIGURE 3.3. The iterations of $\mathcal{N}, \mathcal{S}$ and J on S^1

On the other hand, according to the second assertion of Lemma 3.2 and (3.4), we have $\mathcal{N}_i, \mathcal{S}_i \notin \operatorname{Im}\Phi_G, i = 0, \ldots, n-1$. Together with (3.6), it follows that the number of connected components of $\operatorname{Im}\Phi_G$ exceeds two. This contradicts to the third assertion of Lemma 3.2. Hence, we conclude that $n \leq 2$.

When $n = 1$, we have $\theta = 2m\pi$ and $\rho_\theta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. This rotation leaves $\operatorname{Im}\Phi_G$ fixed.

When $n = 2$, we have $\theta = m\pi$ and $\rho_\theta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. The latter rotation leaves $\operatorname{Im}\Phi_G$ invariant by the first assertion of Lemma 3.2.

Therefore, we complete the proof of Proposition 3.1. □

3.3. **Lipschitz Continuity.** The aim of this subsection is to show that the underlying function of a self-similar graph must be Lipschitz (see Proposition 3.7). To this end, we first cover G by a minimal rectangle R and use the self-similarity of G to obtain a collection of smaller rectangles, each similar to R and with widths less than a given number δ. After projecting them to x-axis, we select out a mimimal cover of a given closed interval, as illustrated in Figure 3.5. The variation of the function on the projection of each rectangle is controlled by the height of those rectangles, which directly yields the Lipschitz continuity of the underlying function.

Proposition 3.7. *Suppose $f : [0,1] \to \mathbb{R}$ continuous. If the graph G of f on $[0,1]$ is self-similar, then f must be a Lipschitz function.*

Let's start with some definitions and notations.

Given a bounded closed interval I and a function $f : I \to \mathbb{R}$, the variation of f on I, is defined as

$$V(f, I) := \max_{x,y \in I} |f(x) - f(y)|. \tag{3.8}$$

We say the graph of one-variable continuous function f on a closed and bounded interval I is framed by a rectangle R, say $[x_1, x_2] \times [y_1, y_2]$, if $I = [x_1, x_2]$ and $y_1 := \min_{x \in I} f(x)$, $y_2 := \max_{x \in I} f(x)$. Furthermore, the height of R is exactly the variation of f on I.

Proof of Proposition 3.7. Since f is continuous and $[0,1]$ is compact, the variation $V(f, [0,1])$, or simply $V(f)$, is finite. We aim to show that given any $\delta > 0$, the inequality $|f(x) - f(y)| \leq 4V(f) \cdot |x - y|$ holds, for any $x, y \in [0,1]$ with $|x - y| = \delta$. By exchanging x and y if necessary, we assume that $x < y$.

Let R be the rectangle framing f on $[0,1]$. For any $n \in \mathbb{N}$ and $\alpha \in [k]^n$, denote by

$$I_\alpha := \pi \circ S_\alpha(R) \tag{3.9}$$

where π is the projection given by $\pi(x_1, x_2) = x_1$.

For each $i \in [k]$, due to $S_i(G) \subset G$, it then follows from Proposition 3.1 that we have $\rho_{\theta_i} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Moreover, $S_i(G)$ is the graph of f on I_i and the rectangle $S_i(R)$ frames the graph of f on I_i. This is illustrated in the Figure 3.4. Obviously $S_i(R)$ is a rectangle with width $|I_i| = r_i$ and height $V(f, I_i) = r_i \cdot V(f)$. Hence, this implies

$$V(f, I_i) = |I_i| \cdot V(f), \text{ for all } i \in [k].$$

Proceeding with the above result for finitely many steps, we obtain that for any $n \in N$ and $\alpha \in [k]^n$,

(1) $S_\alpha(G)$ is the graph of f on I_α;
(2) $S_\alpha(R)$ frames the graph of f on I_α, where $S_\alpha(R)$ is a rectangle with width $|I_\alpha| = r_\alpha$ and height

$$V(f, I_\alpha) = r_\alpha \cdot V(f). \tag{3.10}$$

On the other hand, since $r_1, \ldots, r_n \in (0,1)$, we have $r_{\max} := \max\{r_1, \ldots, r_n\} \in (0,1)$. Thus, we can choose a $n_0 \in \mathbb{N}$ with $r_{\max}^{n_0} \leq \delta$. This implies that for any $\alpha \in [k]^{n_0}$, we have

$$|I_\alpha| = r_\alpha \leq r_{\max}^{n_0} \leq \delta. \tag{3.11}$$

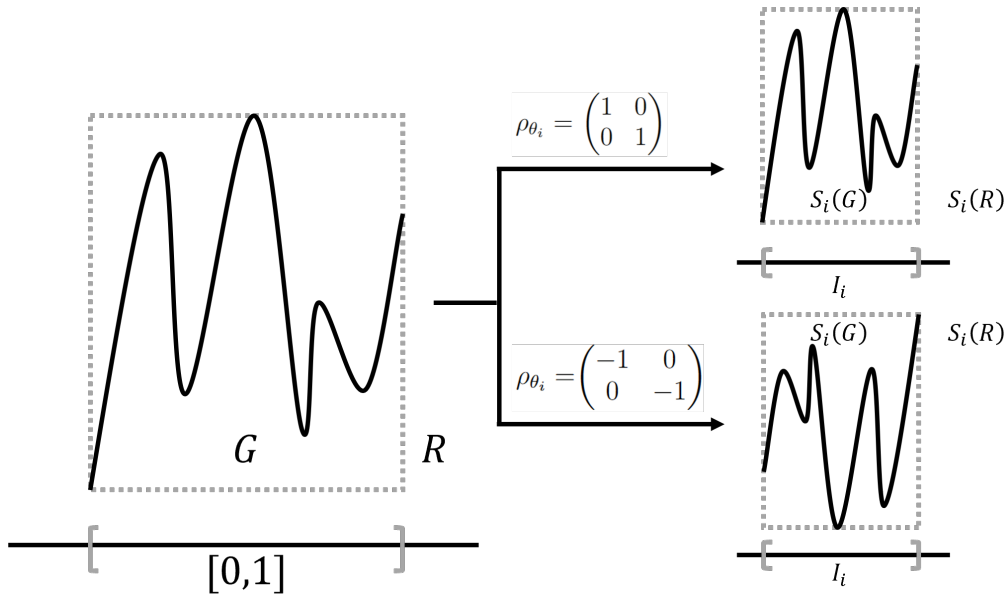

FIGURE 3.4. $S_i(R)$ frames f on I_i

The self-similarity of G indicates that

$$G = \bigcup_{i=1}^{k} S_i(G) = \bigcup_{\alpha \in [k]^{n_0}} S_\alpha(G).$$

Hence G is covered by $\{S_\alpha(R)\}_{\alpha \in [k]^{n_0}}$. Consequently, the interval $[0,1]$ is covered by $\{I_\alpha\}_{\alpha \in [k]^{n_0}}$.

Given the finiteness of $[k]^{n_0}$, we can select a subset of indices $\Lambda \subset [k]^{n_0}$ that meets the following conditions:

(1) $\{I_\alpha\}_{\alpha \in \Lambda}$ forms a cover for $[x, y]$.
(2) $\{I_\alpha\}_{\alpha \in \Lambda}$ is minimal, in the sense that if any $\alpha_0 \in \Lambda$ is removed, $\{I_\alpha\}_{\alpha \in \Lambda \setminus \{\alpha_0\}}$ no longer forms a cover for $[x, y]$.

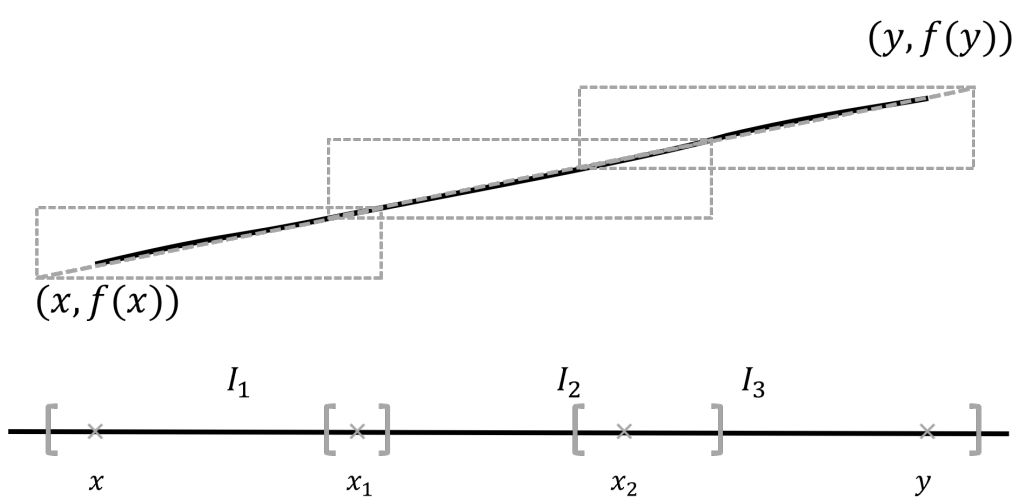

FIGURE 3.5. Minimal cover of $[x, y]$

We order $\{I_\alpha\}_{\alpha\in\Lambda}$ as $\{I_1,\dots,I_m\}$ according to their left endpoints from left to right, as illustrated in Figure 3.5. By condition (2) in our construction, we have $I_j \cap I_{j+2} = \emptyset$ for all $j \in [m-2]$. To see this, suppose for contradiction that there exists some $j_0 \in [m-2]$ such that $I_{j_0} \cap I_{j_0+2}$ is non-empty. Given our ordering, this would imply $I_{j_0+1} \subset I_{j_0} \cup I_{j_0+2}$. In other words, the set $\{I_1,\dots,I_m\}\backslash\{I_{j_0+1}\}$ would still form a cover of $[x,y]$, which contradicts condition (2) in the definition of Λ.

Notice the intervals I_j, where $j \in [m]\backslash\{1,m\}$ and j is odd, are mutually disjoint and their union is contained in $[x,y]$. Therefore, we have

$$\sum_{\substack{j\in[m]\backslash\{1,m\}\\ j \text{ odd}}} |I_j| \le |[x,y]| = \delta.$$

Similarly, we also have

$$\sum_{\substack{j\in[m]\backslash\{1,m\}\\ j \text{ even}}} |I_j| \le \delta.$$

On the other hand, due to (3.11), $|I_1|, |I_m| < \delta$. Hence the total length

$$\sum_{j=1}^{m} |I_j| \le 4\delta. \tag{3.12}$$

Finally, we arbitrarily choose points $x_i \in I_i \cap I_{i+1}, i \in [m-1]$. Consider,

$$\begin{aligned}
|f(x)-f(y)| &\le |f(x)-f(x_1)| + \sum_{i=1}^{m-2} |f(x_i)-f(x_{i+1})| \\
&\quad + |f(x_m)-f(y)| && \text{(by triangle inequality)}\\
&\le V(f,I_1) + \sum_{i=1}^{m-2} V(f,I_{i+1}) + V(f,I_m) && \text{(by (3.8))}\\
&= |I_1|\cdot V(f) + \sum_{i=1}^{m-2} |I_i|\cdot V(f) + |I_m|\cdot V(f) && \text{(by (3.10))}\\
&\le 4\delta V(f). && \text{(by (3.12))}
\end{aligned}$$

Since $\delta > 0$ and $x, y \in [0,1]$ are arbitrarily chosen, we have shown that f is $4V(f)$-Lipschitz. This completes the proof of Proposition 3.7. □

3.4. **Proof of the Main Result.** The aim of this section is to show that the underlying function of a self-similar graph is affine (see Theorem 1.1). To prove this, we demonstrate that any interval $[a,b]$ within $[0,1]$ contains a subinterval $[s,t]$ of comparable length to $[a,b]$ satisfying the condition $\frac{f(t)-f(s)}{t-s} = f(1)-f(0)$ (see Lemma 3.8). By repeatedly applying Lemma 3.8, we can construct a Cantor-like subset within $[a,b]$. This construction, when combined with the fact that f satisfies Lipschitz continuity (see Proposition 3.7), allows us to take a distortion estimate and then deduce that f is affine.

Lemma 3.8. *Let G be the graph of a continuous function f on $[0,1]$. Denote by $\lambda := f(1)-f(0)$. If G is self-similar, then there exists $c \in (0,1)$ such that for any interval $[a,b] \subset [0,1]$, there exists a subinterval $[s,t] \subset [a,b]$ satisfying*

(1) $\frac{a+b}{2} \in [s,t]$;
(2) $c(b-a) \leq t-s \leq \frac{1}{2}(b-a)$;
(3) $\frac{f(t)-f(s)}{t-s} = \lambda$.

Proof of Lemma 3.8. Let's maintain the notations introduced in Proposition 3.7. Take $c := \frac{r_{\min}}{2}$ where $r_{\min} := \min\{r_1, \ldots, r_k\}$. Let's fix an interval $[a,b] \subset [0,1]$. For any integer $n \in \mathbb{N}$, there exists $\alpha \in [k]^n$ such that I_α contains the middle point $\frac{a+b}{2}$, due to the self-smilarity of G. Let n_0 be the smallest integer such that $|I_\alpha| \leq \frac{b-a}{2}$. We show that $I_\alpha = [s,t]$ is the subinterval satisfying three properties mentioned in the statement.

Property (1) and the second inequity in Property (2) are immediate from construction.

For the first inequity in Property (2), we write $\alpha = (\alpha_1, \ldots, \alpha_{n_0}) \in [k]^{n_0}$ and denote by $\alpha' := (\alpha_1, \ldots, \alpha_{n_0-1}) \in [k]^{n_0-1}$. Due to the minimality of n_0, it follows that $|I_{\alpha'}| > \frac{b-a}{2}$. Thus, $|I_\alpha| = r_{\alpha_{n_0}} \cdot |I_{\alpha'}| > r_{\alpha_{n_0}} \cdot \frac{b-a}{2}$, which yields $|I_\alpha| \geq r_{\min} \cdot \frac{b-a}{2} = c(b-a)$.

For Property (3), it suffices to show that if $I_\alpha = [s,t]$, then $\frac{f(t)-f(s)}{t-s} = \lambda$. To see this, we prove the following claim.

Claim. Denote by $\overrightarrow{a} := (a, f(a)) \in G$ where $a \in [0,1]$. For any $n \in \mathbb{N}$ and $\alpha \in [k]^n$, if $\{\overrightarrow{a}, \overrightarrow{b}\} = \{S_\alpha(\overrightarrow{c}), S_\alpha(\overrightarrow{d})\}$, then $\frac{f(b)-f(a)}{b-a} = \frac{f(d)-f(c)}{d-c}$.

Define $\Psi : S^1 \backslash \{\mathcal{N}, \mathcal{S}\} \to \mathbb{R}$ via $(x,y) \mapsto \frac{y}{x}$. It's immediate that Ψ is antipodal symmetric and homogeneous, that is $\Psi(-x,-y) = \Psi(x,y)$ and $\Psi(\mu x, \mu y) = \Psi(x,y)$ for any non-zero $\mu \in \mathbb{R}$.

If $\{\overrightarrow{a}, \overrightarrow{b}\} = \{S_i(\overrightarrow{c}), S_i(\overrightarrow{d})\}$, then

$$\begin{aligned}
&\frac{f(b)-f(a)}{b-a} \\
=&\Psi(\Phi_G(\overrightarrow{a}, \overrightarrow{b})) && \text{(since } \Psi \text{ is homogeneous)} \\
=&\Psi(\Phi_G(S_i(\overrightarrow{c}), S_i(\overrightarrow{d}))) && \text{(since } \Phi_G \text{ is anti-symmetric and } \Psi \text{ is antipodal symmetric)} \\
=&\Psi(\rho_{\theta_i} \circ \Phi_G(\overrightarrow{c}, \overrightarrow{d})) && \text{(by (3.3))} \\
=&\Psi(\Phi_G(\overrightarrow{c}, \overrightarrow{d})) && \text{(by Proposition 3.1 and since } \Psi \text{ is antipodal symmetric)} \\
=&\frac{f(d)-f(c)}{d-c}. && \text{(since } \Psi \text{ is homogeneous)}
\end{aligned}$$

For any $n \in \mathbb{N}$ and $\alpha \in [k]^n$, suppose $\{\overrightarrow{a}, \overrightarrow{b}\} = \{S_\alpha(\overrightarrow{c}), S_\alpha(\overrightarrow{d})\}$. By applying the above result for n times, we have $\frac{f(b)-f(a)}{b-a} = \frac{f(d)-f(c)}{d-c}$, which completes the proof of the claim.

Now, let's carry on our proof of Lemma 3.8. Since $\{\overrightarrow{s}, \overrightarrow{t}\} = \{S_\alpha(\overrightarrow{0}), S_\alpha(\overrightarrow{1})\}$, we have $\frac{f(t)-f(s)}{t-s} = \frac{f(1)-f(0)}{1-0} = \lambda$ by the claim. This concludes our proof of Lemma 3.8. □

Now we are ready to prove our main result, Theorem 1.1.

Proof of Theorem 1.1. " $\Rightarrow$ " : To show f is affine, it suffices to show that for any interval $[a,b] \subset [0,1]$, we have $\frac{f(b)-f(a)}{b-a} = \lambda$.

Given an arbitrary interval $[a,b]$, we construct a Cantor-like subset of $[a,b]$ inductively in the following way.

For initial step, there exists a subinterval of $[a, b]$, say $[a_1^1, b_1^1]$, satisfying three properties in Lemma 3.8. We define $C_1 := [a, b] \backslash (a_1, b_1)$. This set is non-empty due to the first two properties in Lemma 3.8.

Suppose we have defined C_k. Notice there are 2^k intervals in C_k. For stage $k+1$, we apply Lemma 3.8 on each of these 2^k intervals, obtaining subintervals $[a_1^k, b_1^k], \ldots, [a_{2^k}^k, b_{2^k}^k] \subset C_k$. We then define $C_{k+1} := C_k \backslash \bigcup_{i=1}^{2^k} (a_i^k, b_i^k)$. This set is also non-empty by the first two properties in Lemma 3.8.

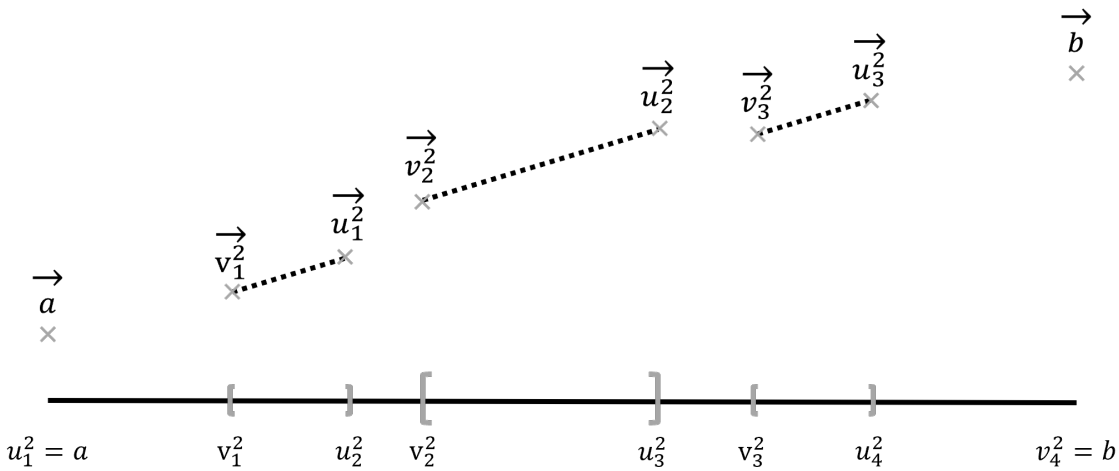

FIGURE 3.6. Illustration of the set C_2 in the second step

In the set C_k, which contains 2^k intervals, we denote the right and left endpoints of the i-th interval as u_i and v_i respectively. Thus, the intervals in C_k can be represented as $[u_1^k, v_1^k], [u_2^k, v_2^k], \ldots, [u_{2^k}^k, v_{2^k}^k]$, as depicted in Figure 3.6. Next, we define the total length of C_k as $|C_k| := \sum_{i=1}^{2^k} (v_i^k - u_i^k)$. By invoking Property (2) in Lemma 3.8, we deduce that $|C_{k+1}| \leq (1-c)|C_k|$. Combined this with the fact that $|C_1| \leq (1-c)(b-a)$, it follows that

$$|C_k| \leq (1-c)^k (b-a). \tag{3.13}$$

By Proposition 3.7, f is Lipschitz, say it is L-Lipschitz. Given any $k \in \mathbb{N}$, we have

$$|f(v_i^k) - f(u_i^k)| \leq L|v_i^k - u_i^k|, \text{ for all } i \in [2^k]. \tag{3.14}$$

On the other hand, by Property (3) in Lemma 3.8, it follows that

$$f(u_{i+1}^k) - f(v_i^k) = \lambda(u_{i+1}^k - v_i^k), \text{ for all } i \in [2^k - 1]. \tag{3.15}$$

For all $k \in \mathbb{N}$, we can deduce that

$$\begin{aligned}
&|f(b) - f(a) - \lambda(b-a)| \\
=&|\sum_{i=1}^{2^k}(f(v_i^k) - f(u_i^k)) + \sum_{i=1}^{2^k-1}(f(u_{i+1}^k) - f(v_i^k)) - \lambda(b-a)| \\
=&|\sum_{i=1}^{2^k}(f(v_i^k) - f(u_i^k)) + \sum_{i=1}^{2^k-1}\lambda(u_{i+1}^k - v_i^k) - \lambda(b-a)| && \text{(by (3.15))} \\
\leq&\sum_{i=1}^{2^k}|f(v_i^k) - f(u_i^k)| + |\lambda(b-a-|C_k|) - \lambda(b-a)| \\
\leq&(L+|\lambda|)|C_k| && \text{(by (3.14))} \\
\leq&(L+|\lambda|)(1-c)^k(b-a). && \text{(by (3.13))}
\end{aligned}$$

Since $c = \frac{r_{\min}}{2} < 1$ and k is arbitrary, we conclude that $f(b) - f(a) - \lambda(b-a) = 0$ for any interval $[a,b] \subset [0,1]$. Therefore, we have proven that f is affine, which proves the forward implication.

" $\Leftarrow$ " : Conversely, suppose f is an affine function. Similitudes $S_1(x) = \frac{1}{2}x + (0, \frac{f(0)}{2})$ and $S_2(x) = \frac{1}{2}x + (\frac{1}{2}, \frac{f(1)}{2})$ satisfies $G = S_1(G) \cup S_2(G)$. Hence, G is self-similar. □

As a direct corollary of Theorem 1.1, we have

Corollary 3.9. *The graph of a continuous but non-affine function on a closed and bounded interval is not self-similar.*

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